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Steg i rent trykk

$f_y := 460\ \text{MPa}$

$\varepsilon := \sqrt{\frac{235\ \text{MPa}}{f_y}} = 0.7148$

IPE450

h	b	tw	tf	r
450	190	9.4	14.6	21

$h := 450\ \text{mm}$ $\quad b := 190\ \text{mm}$

$t_w := 9.4\ \text{mm}$ $\quad t_f := 14.6\ \text{mm}$ $\quad r := 21\ \text{mm}$

$b_p := h - 2 \cdot t_f - 2 \cdot r = 378.8\ \text{mm}$

$t_p := t_w = 0.0094\ \text{m}$

IPE240

h	b	tw	tf	r
240	120	6.2	9.8	15

Pure compression - internal part

$\psi := 1.0$ sigma 1 = sigma 2 - no bending

$k_\sigma := 4$ bottom of table 4.1

$\frac{b_p}{t_p} = 40.2979$

$\lambda_{p_} := \frac{\frac{b_p}{t_p}}{28.4 \cdot \varepsilon \cdot \sqrt{k_\sigma}} = 0.9926$

$$\rho_{internal} := \begin{cases} 1.0 & \text{if } \lambda_{p_} \le 0.5 + \sqrt{0.085 - 0.055 \cdot \psi} \\ \min\left(\left[\frac{\lambda_{p_} - 0.055 \cdot \psi}{\lambda_{p_}^{\ 2}}\right]\right) & \text{else} \end{cases} = 0.9516$$

$b_{eff} := \rho_{internal} \cdot b_p = 360.4747\ \text{mm}$ remaining area

$b_{removed} := b_p - b_{eff} = 18.3253\ \text{mm}$ $\quad A_{gross} := 9882\ \text{mm}^2$

$A_{removed} := b_{removed} \cdot t_w = 172.2578\ \text{mm}^2$ $\quad A_{eff} := A_{gross} - A_{removed} = 9709.7422\ \text{mm}^2$

$e_{z.removed} := 0\ \text{mm}$

$I_{y.gross} := 337.4 \cdot 10^6\ \text{mm}^4$ $\quad I_{removed} := \frac{b_{removed}^{\ 3} \cdot t_w}{12} + A_{removed} \cdot e_{z.removed}^{\ 2} = 4820.582\ \text{mm}^4$

$I_{y.eff} := I_{y.gross} - \frac{b_{removed}^{\ 3} \cdot t_w}{12} - I_{removed} = 337.3904\ 10^6\ \text{mm}^4$

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100x3 CSHS S460

Internal pure compression elements - alle 4 sider

$b := 100\ \text{mm}$ $\quad t_p := 3\ \text{mm}$ $\quad f_y := 460\ \text{MPa}$

$r_o := 6\ \text{mm}$

$\varepsilon := \sqrt{\frac{235\ \text{MPa}}{f_y}} = 0.7148$

$b_p := b - 2 \cdot r_o = 88\ \text{mm}$

$\psi := 1.0$ sigma 1 = sigma 2 - no bending

$k_\sigma := 4$ bottom of table 4.1

$$\lambda_{p_} := \frac{\frac{b_p}{t_p}}{28.4 \cdot \varepsilon \cdot \sqrt{k_\sigma}} = 0.7225 \qquad 0.5 + \sqrt{0.085 - 0.055 \cdot \psi} = 0.6732$$

$$\rho_{internal} := \text{if } \lambda_{p_} \le 0.5 + \sqrt{0.085 - 0.055 \cdot \psi} = 1$$
$$1.0$$
$$\text{else}$$
$$\min\left(\begin{bmatrix} 1.0 \\ \frac{\lambda_{p_} - 0.055 \cdot \psi}{\lambda_{p_}^{\ 2}} \end{bmatrix}\right) \qquad \frac{\lambda_{p_} - 0.055 \cdot \psi}{\lambda_{p_}^{\ 2}} = 1.2787$$

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180x4 CSHS S460

Internal pure compression elements - alle 4 sider

$b := 180\ \text{mm}$ $\quad t_p := 4\ \text{mm}$ $\quad f_y := 460\ \text{MPa}$

$r_o := 8\ \text{mm}$ $\quad \varepsilon := \sqrt{\frac{235\ \text{MPa}}{f_y}} = 0.7148$

$b_p := b - 2 \cdot r_o = 164\ \text{mm}$

$\psi := 1.0$ sigma 1 = sigma 2 - no bending

$k_\sigma := 4$ bottom of table 4.1

$$\lambda_{p_} := \frac{\frac{b_p}{t_p}}{28.4 \cdot \varepsilon \cdot \sqrt{k_\sigma}} = 1.0099 \qquad 0.5 + \sqrt{0.085 - 0.055 \cdot \psi} = 0.6732$$

$$\rho_{internal} := \text{if } \lambda_{p_} \le 0.5 + \sqrt{0.085 - 0.055 \cdot \psi} = 0.9363$$
$$1.0$$
$$\text{else}$$
$$\min\left(\left[\begin{array}{c} 1.0 \\ \frac{\lambda_{p_} - 0.055 \cdot \psi}{\lambda_{p_}^{\ 2}} \end{array}\right]\right)$$

$b_{eff} := \rho_{internal} \cdot b_p = 153.5476\ \text{mm}$ remaining area

$b_{removed} := b_p - b_{eff} = 10.4524\ \text{mm}$ centric on plate

$A_{p.i.removed} := b_{removed} \cdot t_p = 41.8097\ \text{mm}^2$

$A_{removed.tot} := 4 \cdot A_{p.i.removed} = 167.2387\ \text{mm}^2$

$A_{gross} := 2775\ \text{mm}^2$

$A_{eff} := A_{gross} - A_{removed.tot} = 2607.7613\ \text{mm}^2$ $\quad I_{y.gross} := 14.22 \cdot 10^6\ \text{mm}^4$

$$I_{y.removed.vertical} := 2 \cdot \left(\frac{b_{removed}^{\ 3} \cdot t_p}{12} + (0\ \text{mm})^2 \cdot A_{p.i.removed}\right) = 761.3053\ \text{mm}^4$$

$$I_{y.removed.horizontal} := 2 \cdot \left(\frac{t_p^{\ 3} \cdot b_{removed}}{12} + A_{p.i.removed} \cdot \left(\frac{b}{2} - \frac{t_p}{2}\right)^2\right) = 6.4766 \cdot 10^5\ \text{mm}^4$$

$$I_{y.eff} := I_{y.gross} - I_{y.removed.vertical} - I_{y.removed.horizontal} = 1.3572 \cdot 10^7\ \text{mm}^4$$

$$\frac{I_{y.eff}}{I_{y.gross}} = 0.9544$$

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